Gravitation

Science 1 (Class 10)

What is Force?

Force is a push or pull that can change how things move or look.

- Can change an object's shape
- Can change speed of movement
- Can change direction of motion
- Happens all around us every day

Examples: Pushing a door open, kicking a soccer ball, pulling a wagon

Understanding Motion

Motion Definition

When an object changes its position with time, it's in motion

Rest vs Motion

Objects can be at rest (not moving) or in motion (moving)

Real Examples

A moving car, a walking person, a flying bird

How Force and Motion Connect

The Connection

Force causes motion - without force, objects stay still or keep moving in the same way.

- More force creates more change in motion
- Force changes speed or direction
- Stopping also needs force

> **Example:** Kicking a ball harder makes it go farther

Circular Motion

When an object moves in a circular path, it's called circular motion. The direction keeps changing, even if speed stays the same.

Electric Fan

Fan blades rotate in a circle

Earth's Rotation

Our planet spins around its axis

Ferris Wheel

Circular motion at amusement parks

Centripetal Force

What is it?

The force that pulls objects toward the center when they move in a circle.

- Always points toward the center
- Keeps objects in circular path
- Without it, objects would fly away

Example: When you swing a stone tied to a string, the string pulls the stone toward your hand (the center)

Gravitation

Force of Attraction

Every object with mass attracts every other object

Everywhere in Universe

Gravity works throughout space, not just on Earth

Common Examples

Apple falling from tree, we stay on Earth's surface

Newton's Universal Law of Gravitation

Every object in the universe attracts every other object with a force called gravity.

Key Points

- Bigger mass = stronger force
- Smaller distance = stronger force
- Depends on both mass and distance
- Works everywhere in space

Example: Earth's large mass attracts us and keeps us on the ground

Kepler's Laws of Planetary Motion

Johannes Kepler discovered three laws that describe how planets move around the Sun.

1

Law 1: Elliptical Orbits

Planets move in oval-shaped paths (ellipses) around the Sun, not perfect circles

2

Law 2: Changing Speed

Planets move faster when closer to the Sun and slower when farther away

3

Law 3: Distance and Time

The time it takes to orbit depends on how far the planet is from the Sun

Key Takeaways

Force Affects Motion

Force can change shape, speed, and direction of objects

Circular Needs Center Force

Centripetal force keeps objects moving in circles

Gravity Pulls Everything

Every object attracts every other object in the universe

Scientists Explained Space

Newton and Kepler helped us understand how planets move